

Final Task – HR Database System

This final task is designed for students to create a simple HR database, insert sample data, and write SQL SELECT queries to test their understanding of database design, relationships, and querying.

1. Objective

- Design a simple HR database schema.
- Create the required tables and relationships.
- Insert sample data using SQL INSERT statements.
- Write a set of SELECT queries to retrieve meaningful information.

2. Required Tools

Use PostgreSQL, MySQL, or any SQL-based database system along with pgAdmin, MySQL Workbench, or DBeaver.

3. Database Schema

Students must create the following tables and relationships:

Table: departments

- - department_id INT PRIMARY KEY AUTO_INCREMENT
- - name VARCHAR(100) NOT NULL UNIQUE
- - location VARCHAR(100)

Table: jobs

- - job_id INT PRIMARY KEY AUTO_INCREMENT
- - title VARCHAR(100) NOT NULL UNIQUE
- - min_salary DECIMAL(10,2) CHECK(min_salary >= 0)
- - max_salary DECIMAL(10,2) CHECK(max_salary >= min_salary)

Table: employees

- - employee_id INT PRIMARY KEY AUTO_INCREMENT
- - first_name VARCHAR(50) NOT NULL
- - last_name VARCHAR(50) NOT NULL
- - email VARCHAR(120) UNIQUE
- - phone VARCHAR(40)
- - hire_date DATE NOT NULL
- - department_id INT REFERENCES departments(department_id)
- - job_id INT REFERENCES jobs(job_id)
- - manager_id INT REFERENCES employees(employee_id)

Table: salaries

- - salary_id INT PRIMARY KEY AUTO_INCREMENT
- - employee_id INT REFERENCES employees(employee_id)

- - start_date DATE NOT NULL
- - end_date DATE
- - salary DECIMAL(10,2) NOT NULL

Table: leaves

- - leave_id INT PRIMARY KEY AUTO_INCREMENT
- - employee_id INT REFERENCES employees(employee_id)
- - start_date DATE NOT NULL
- - end_date DATE NOT NULL
- - leave_type VARCHAR(30) CHECK(leave_type IN ('Annual','Sick','Unpaid','Other'))
- - status VARCHAR(20) CHECK(status IN ('Approved','Pending','Rejected'))

4. Relationships

- One Department has many Employees
- One Job has many Employees
- One Employee can have many Salaries
- One Employee can have many Leaves
- Employees table has self-referencing relationship for manager_id

5. Required Tasks

1. Create all the tables in the order given above.
2. Insert at least:
 - 3 departments
 - 6 job titles
 - 7 employees (3 managers + 4 staff)
 - 1 current salary per employee
 - 4 leave records
3. Apply all primary key, foreign key, and check constraints.
4. Test your data by running SELECT queries below.

6. SQL SELECT Queries

1. 1) List all employees with department and job title.
2. 2) Show all employees in IT department only.
3. 3) Count the number of employees per department.
4. 4) Average current salary per job title.
5. 5) Highest current salary with employee name and department.
6. 6) Employees without manager.
7. 7) Employees with their manager names.
8. 8) List of current salaries with start date.
9. 9) Employees with approved leaves in 2025.
10. 10) Total approved leave days per employee in 2025.
11. 11) Number of employees per job title.
12. 12) Employees without any leaves in 2025.
13. 13) Average salary per department.

14. 14) Employees with salary above department average.
15. 15) Employees hired after 2023-01-01.

7. Submission Requirements

- SQL file with CREATE, INSERT, and SELECT statements.
- 3–5 screenshots showing your tables and query results.
- Optional: PDF or Word summary file with notes.

8. Evaluation Rubric (100 points)

- Schema and constraints – 35 points
- Sample data completeness – 20 points
- Correctness of SELECT queries – 35 points
- Organization and presentation – 10 points